When Self-Supervision Transfers: Quantifying the Laparoscopic–Robotic Domain Shift

***Abstract*— Minimally invasive surgery encompasses laparoscopic and robotic surgery. Self-supervised learning (SSL) is the dominant paradigm for learning visual representations from unlabelled surgical videos, yet virtually all large-scale efforts target laparoscopic data. Whether the resulting representations transfer to robotic surgery—where optics, viewpoints, and instrument appearance differ markedly—remains an open question.**

We present the first systematic study of this laparoscopic–robotic domain gap. We curate one of the largest purely robotic video collections reported to date (10 datasets; 43,425 clips; 499.4 h; ∼71.3M frames) and a robotic cholecystectomy benchmark with phase labels mapped to the Cholec80 ontology, collected across more than 20 surgeons. We describe four experiments using EndoViT [1] (endoscopic Vision Transformer) with an SPR head (phase classifier) to provide a deeper understanding of the gap: (E1) cross-domain evaluation reveals severe bidirectional transfer collapse (both below majority-class guessing); (E2) fine-tuning under frozen, partially unfrozen (one additional linear layer), and fully unfrozen backbone regimes; (E3) progressive laparoscopic data injection improves minority-phase robustness at moderate ratios but causes transient negative transfer at high ratios; (E4) robotic-only vs. mixed Mask Auto Encoder pretraining converge under full fine-tuning but diverge under constrained adaptation, with different precision–recall trade-offs.

Our results frame transferability as a three-axis design problem—pretraining modality, adaptation depth, and data composition—and provide actionable guidance for deploying Self-supervised learning backbones in robotic surgical workflows.

I. Introduction

Self-supervised learning (SSL) has become the dominant strategy for leveraging the vast amounts of unlabelled surgical video that exist today. By learning visual representations without manual annotations, SSL promises scalable feature extractors that can be fine-tuned for downstream tasks such as phase recognition, tool detection, and action triplet prediction. A growing body of work now demonstrates the effectiveness of this paradigm in the surgical domain: SelfSupSurg [2] benchmarks prominent SSL paradigms across multiple laparoscopic datasets; EndoViT [1] pretrains vision transformers on the Endo700k endoscopic collection; SurgVISTA [3] extends pretraining to the video level with spatiotemporal objectives; and foundation models such as Endo-FM [4], EndoMamba [5], and Surg-3M/SurgFM [6] push both scale and generality further. A parallel line of work leverages natural language as supervision to improve transfer and zero-shot capability, including SurgVLP [7], HecVL [8], and VidLPRO [9].

Despite this progress, virtually all large-scale surgical SSL efforts have been trained on—and evaluated against—*laparoscopic* data. *Laparoscopy* denotes manual minimally invasive surgery performed through small ports under a rigid endoscopic camera, whereas *robotic-assisted* surgery uses a teleoperated robotic platform with articulated instruments and typically more stable, system-controlled endoscopic viewpoints, yielding different tool appearances, motion statistics, and visual cues.

Self-supervised models tailored to *robotic* surgery remain comparatively scarce, with only isolated efforts [10], [11] exploring robotic-specific pretraining. More critically, it is still unclear whether representations learned from laparoscopic video transfer reliably to robotic settings, or whether domain-specific differences in optics, viewpoint, instrument appearance, and image framing introduce a distribution shift that undermines transfer.

In this work, we systematically study the **transferability of Mask Autoencoders (MAE) [12] self-supervised representations** [13] **between laparoscopic and robotic surgery**, while contextualising MAE against other widely used SSL/backbone families such as DINO-style distillation [14], ResNet-based representation learning [15], and surgical SSL frameworks including EndoSSL [16] and SelfSupSurg [2].

To enable this study, we first curate one of the largest purely robotic multi-dataset collections reported in the literature, **OmniRAS**, by merging 10 public and in-house datasets (43,425 clips; 499.4 h; ∼71.3M frames). The dataset exceeds SurgToolLoc2022 alone in clip count (24,686 clips) and includes our own private dataset, **OmniRAS-Private**, collected across more than 20 operating rooms. The corpus is concentrated at 640×360 and 1920×1080 resolutions, with average clip duration ∼41 s and frame rates spanning 2–60 fps (39.7 fps weighted average). We describe the full curation pipeline that includes automated laparoscopic/robotic separation, clip boundary detection, and interactive cleaning. A subset of OmniRAS-Private, **OmniRAS-PR** has been subsequently annotated with frame-level surgical phase labels mapped to the Cholec80 label space, creating a controlled cross-modality benchmark with identical classes and metrics.

This study focuses on two backbones: EndoViT SPR [1] and our own robotic MAE-pretrained ViT. We use Cholec80 as the laparoscopic baseline and our annotated OmniRAS-Private subset as the robotic-side benchmark. Using these settings, we conduct a four-pronged transferability study (experiments E1–E4). In **E1** we fine-tune the laparoscopic-pretrained backbone on Cholec80 and test directly on OmniRAS-PR to establish whether cross-domain evaluation degrades performance. In **E2** we fine-tune on OmniRAS-PR under frozen, partially unfrozen (one additional linear layer), and fully unfrozen backbone regimes to diagnose whether the

gap is shallow (classifier only) or deep (representation-level). In **E3** we fine-tune on OmniRAS-PR while progressively injecting Cholec80 laparoscopic data, quantifying how much cross-modality exposure is needed to bridge the gap. Finally, in **E4** we perform full MAE pretraining of EndoViT on OmniRAS-Private (14 epochs) before OmniRAS-PR fine-tuning, comparing two pretraining data configurations: robotic corpus only vs. robotic corpus plus the laparoscopic subset of SurgeNet.

Our results reveal a pronounced domain shift: laparoscopic-pretrained features transfer poorly to robotic data, the gap is situated deep in the representation (not just the classifier head), and progressive cross-modality injection can partially but not fully bridge it. These findings motivate modality-aware pretraining strategies for surgical SSL.

II. Data

Three data resources underpin our study. The first is a large-scale, purely robotic video corpus (OmniRAS) assembled for self-supervised pretraining (Section II-A), for which we describe the full curation pipeline, including provenance, modality separation, and clip cleaning. The second is an annotated phase-recognition subset of OmniRAS (OmniRAS-PR) used for supervised fine-tuning and evaluation (Section II-B). Third, we use Cholec80 [17] as the laparoscopic reference dataset for baseline training and cross-modality transfer evaluation.

A. OmniRAS dataset

1) Dataset Characteristics: We merge 10 robotic-surgery datasets with substantial heterogeneity in scale and capture properties: EndoVis15 [18], JIGSAWS [19], MICCAI 2017 [20], MICCAI-EndoSeg [21], SurgToolLoc2022 [22], SurgVisDom (real and VR splits) [23], SurgeNet [24], CRCD [25], and the OmniRAS-Private. The OmniRAS-Private dataset is our in-house robotic surgery dataset, collected across more than 20 surgeons in different domains of expertise, which substantially increases diversity in camera viewpoints, procedures, and acquisition conditions. Importantly, every video in OmniRAS is *purely robotic* (robot-assisted endoscopic views) and does not include any laparoscopic-only recordings, meaning that the datasets composed of both robotic-assisted and laparoscopic videos (SurgeNet, OmniRAS-Private) have been processed to extract and separate robotic subsets. Resolution of the videos in the dataset is concentrated at 640×360 (24,906 clips, predominantly SurgToolLoc2022) and 1920×1080 (11,583 clips, predominantly OmniRAS). Less frequent formats include 1280×720, 640×480, and 720×480. Average clip duration is $\sim$41 s; frame rate spans 2–60 fps (39.7 fps weighted average). To the best of our knowledge, this is one of the largest purely robotic multi-dataset collection reported in the literature in terms of clip count, almost doubling the size of SurgToolLoc2022 (24,686 clips) [22].

Despite this large-scale aggregation, source videos are heterogeneous and may contain edited content not suitable for representation learning. In particular, some recordings (e.g., in

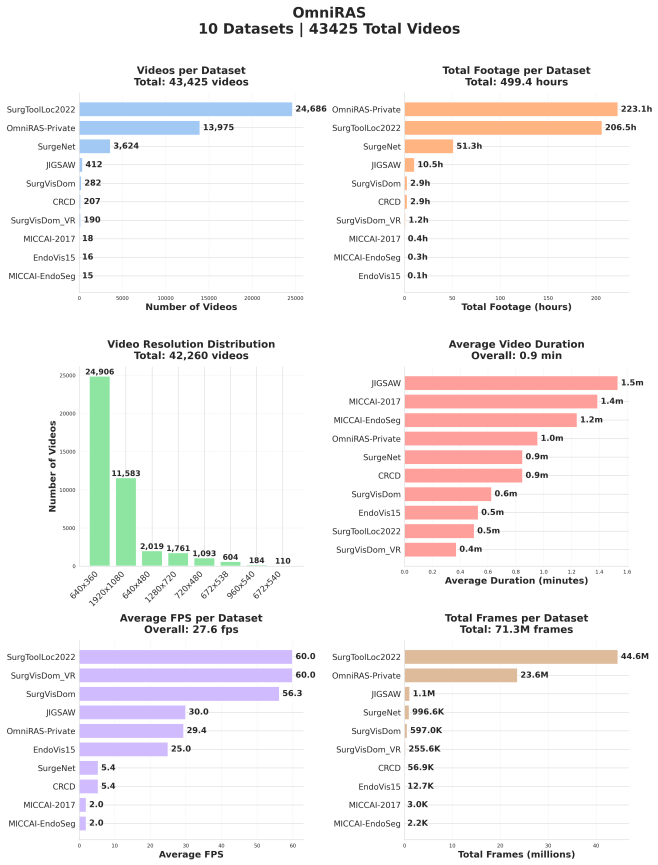

Fig. 1. Provenance statistics of the purely robotic pretraining corpus (video count, total duration, resolution distribution, average clip duration, average FPS, total frames, and storage size).

SurgeNet) are compilation videos that concatenate sequential procedures without explicit modality boundaries, and include non-operative segments such as text slides, comment screens, and animated transitions. This motivates the next two curation steps: (a) laparoscopic/robotic separation; and (b) clip division and cleaning. An additional outcome of this process is a reliable laparoscopic subset, which we later use in the mixed-modality pretraining setting of E4 (IV-D).

2) Laparoscopic/Robotic Separation: We first apply a heuristics based on the black-pixel-per-row profile of each frame: laparoscopic recordings exhibit a circular vignette with pronounced border peaks, whereas robotic frames fully fill the robotic frames (Figure 2). Because this heuristic is sensitive to lighting and exposure, we subsequently train a YOLO11m-cls [26] binary classifier (ImageNet init, 41k images, 80/20 split) that converges to $\sim$99.7% accuracy after $\sim$35 epochs, confirming the two modalities are visually highly discriminative.

3) Clip Division and Cleaning: An error of 0.3%, even if small, can lead to a consequential injection of laparoscopic images into the OmniRAS dataset. Additionally, merged recordings may contain non-surgical content (slides, transitions) or concatenated procedures. To solve this, we segment them using DINOv2 ViT-L/14 [27] frame embeddings: a sliding-window cosine-distance signal $D_w = 1 - \cos(M_w, M_{w+W})$

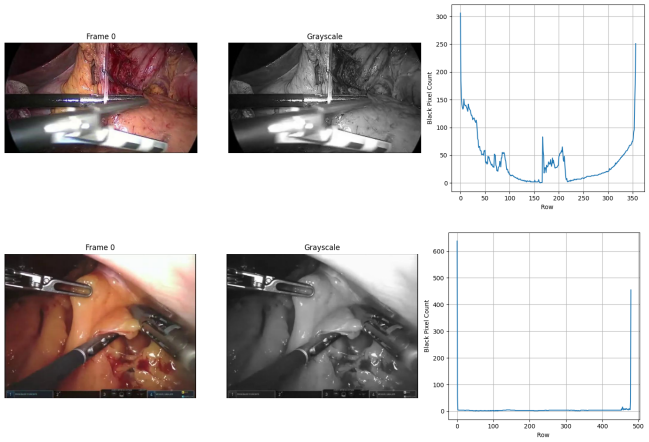

Fig. 2. Black-pixel-per-row profiles for laparoscopic (top, circular field-of-view with borders) and robotic (bottom, full-frame) images.

highlights abrupt content changes, and boundary candidates are produced by PELT [28] change-point detection and peak finding. A lightweight Transformer encoder (4 layers, 8 heads, d_{model}=128) trained on manually accepted boundaries then proposes per-frame boundary probabilities that are verified by a human annotator before final clip splitting.

B. OmniRAS-PR Phase-Recognition Dataset

We build a supervised phase-recognition training-subset OmniRAS-PR, as a dedicated subset separated from OmniRAS, containing only cholecystectomy procedures, manually annotated by surgeons (that we are aiming to release publicly).

For this study, we focus exclusively on phase recognition as a comparatively tool-agnostic task for quantifying cross-modality transfer. In contrast, tool segmentation is inherently tool-dependent, and action triplet recognition is similarly tied to specific instruments and their interactions, which can confound conclusions about modality shift.

1) Phase Ontology and Label Mapping: Because OmniRAS-PR annotations are more fine-grained than the Cholec80 ontology, we define an explicit mapping (Table I) from OmniRAS-PR phases to the closest Cholec80 label. This harmonization enables controlled cross-modality evaluation under identical classes and metrics; it does not replace the richer OmniRAS-PR ontology. Our annotations are more fine-grained and anatomically/action-specific (e.g., anterior vs posterior Calot opening; cystic artery vs duct; clip vs divide), reducing the label ambiguity inherent in Cholec80's coarser, aggregated phases.

2) Dataset Construction: After Cholec80 remapping, 1,166 frames labelled *specimen retrieval* (no Cholec80 equivalent) are removed, leaving **788,387 frames** across five Cholec80 phases. One video is removed by a class-diversity filter ($\geq$50% of phases required), yielding **41 videos** (Table II).

The distribution exhibits a 79.9-fold class imbalance (max/min): *CalotTriangleDissection* (43.3%) and *GallbladderDissection* (44.0%) account for 87.3% of frames, while *GallbladderRetraction* (0.55%) and *Preparation* (1.91%) are extremely rare. To mitigate this, minority classes are upsampled to the median count (16,916) using sampling with augmentation (flips, distortions, blur, noise, dropout, photometric shifts), adding 20,594 copies and yielding **500,915 training frames**.

TABLE I

MAPPING FROM OMNIRAS-PR PHASE LABELS TO CHOLEC80 PHASES FOR CROSS-DATASET EVALUATION.

OmniRAS-PR phase	Cholec80 phase
None	Preparation
Retraction of the gallbladder neck	GallbladderRetraction
Opening the anterior peritoneal layer of the triangle of Calot	CalotTriangleDissection
Opening the posterior peritoneal layer of the triangle of Calot	CalotTriangleDissection
Isolation of the cystic artery	CalotTriangleDissection
Isolation of the cystic duct	CalotTriangleDissection
Clipping of the cystic artery	ClippingCutting
Clipping of the cystic duct	ClippingCutting
Division of the cystic artery	ClippingCutting
Division of the cystic duct	ClippingCutting
Dissection of the gallbladder from the liver	GallbladderDissection

TABLE II

OMNIRAS-PR DATASET STATISTICS AFTER CHOLEC80 PHASE REMAPPING (VIDEO-DISJOINT SPLITS, 30 FPS SEED 1665) — TRANSPOSED.

Phase	All (788,387 fr)	Train (28 vids)	Val (6 vids)	Test (7 vids)
Prep.	15,034 (1.91%)	9,887 (2.06%)	2,409 (1.83%)	2,738 (1.55%)
GB Retr.	4,342 (0.55%)	3,351 (0.70%)	381 (0.29%)	610 (0.35%)
Calot Diss.	341,399 (43.30%)	203,360 (42.34%)	72,225 (54.72%)	65,814 (37.38%)
Clip./Cut.	80,564 (10.22%)	31,855 (6.63%)	13,517 (10.24%)	35,192 (19.99%)
GB Diss.	347,048 (44.02%)	231,868 (48.27%)	43,447 (32.92%)	71,733 (40.74%)
Total frames	**788,387 (100.00%)**	**480,321 (60.92%)**	**131,979 (16.74%)**	**176,087 (22.34%)**

III. EXPERIMENTAL SETUP

We perform four experiments (E1–E4) that progressively isolate the source of the laparoscopic–robotic domain gap.

Across E1–E4, we use a ViT-Base/16 backbone pretrained with an MAE self-supervised objective. For the laparoscopic initialisation, we use the EndoViT-SPR checkpoint (pretrained on Endo700k; `endovit_SPR.pth` [1]); for the robotic initialisation, we use our MAE-pretrained OmniRAS checkpoints as specified in each subsection (III-A,III-B,III-C,III-D). Cholec80 retains its native phase labels, while OmniRAS-PR is mapped to the Cholec80 phase ontology (Table I) so that classes, metrics, and evaluation are consistent across data sources.

Training configuration. Unless otherwise noted: 224×224 inputs, batch size 256, learning rate 5×10^{-5}, weight decay 0.1, 14 epochs. For E2 specifically, we use a shorter 4-epoch fine-tuning schedule. Experiments use three different seeds, 12 workers, 1 NVIDIA RTX 5090.

A. E1: Cross-Domain Performance (Cholec80 ↔ OmniRAS-PR)

We fine-tune EndoViT SPR on Cholec80 (laparoscopic, all layers unfrozen) and evaluate directly on the OmniRAS-PR test set. This measures how well a fully laparoscopic pipeline generalises to robotic data without any robotic exposure during training. We also run the opposite direction: fine-tune

our robotic-pretrained model on the OmniRAS-PR phase-recognition dataset and evaluate on Cholec80, to test whether transfer failure is symmetric across modalities.

B. E2: Depth of Fine-Tuning

Starting from the same laparoscopic-pretrained checkpoint, we fine-tune on OmniRAS-PR under three regimes: (i) *frozen backbone*, where the ViT encoder is fixed and only the task head is updated, and (ii) *partially unfrozen* (*linear layer*), where one additional linear layer is unfrozen together with the task head, and (iii) *full fine-tuning*, where all parameters are updated. A monotonic improvement from frozen to linear-layer to full fine-tuning would indicate that the domain shift is situated deep in the representation, not merely in the classifier. If the linear-layer regime closes most of the gap, the mismatch is shallower; if full fine-tuning remains clearly better, deeper representation adaptation is required.

C. E3: Finetuning with Mixed Data

This experiment aims to demonstrate that augmenting a robotic surgical dataset with procedure-equivalent laparoscopic frames increases data diversity and enhances fine-tuning performance.

For E3, we start from a robotic MAE checkpoint pretrained on OmniRAS for 4 epochs, then fine-tune on OmniRAS-PR while progressively injecting Cholec80 laparoscopic data at fractions of the Cholec80 training set. Crucially, Cholec80 carries a more balanced class distribution, so injection also attenuates OmniRAS-PR's long-tail imbalance (Figure 3). This balancing effect has a much larger relative impact on minority classes (e.g., raising GB retraction from $\leq 1\%$ to $\sim 3.6\%$) than on majority classes (e.g., raising Calot dissection from 45% to 47.9%). All hyperparameters are held constant across fractions to isolate the effect of data composition. We intentionally do not continue MAE pretraining to convergence here (unlike E4), so this experiment isolates the effect of injection itself rather than gains from a fully matured pretraining stage.

This injection curve quantifies how much laparoscopic data is required to recover or exceed the OmniRAS-PR-only baseline, whether a saturation point exists beyond which additional laparoscopic data yields diminishing returns, and whether even small cross-modality exposures already provide a meaningful boost.

D. E4: Pretraining with Mixed Data

As a complementary experiment, we pretrain EndoViT from scratch with a classic MAE objective on the full robotic pretraining corpus for 14 epochs (in contrast to the intentionally incomplete 4-epoch checkpoint used in E3), and then fine-tune for phase recognition. We evaluate two pretraining data configurations: (i) robotic corpus only, and (ii) robotic corpus plus the laparoscopic subset of SurgeNet. Both components are defined in Section II-A. In the mixed-modality setting, the laparoscopic injection corresponds to 245,291 frames from SurgeNet-LP. Comparing this robotic-pretrained backbone against the original laparoscopic-pretrained one

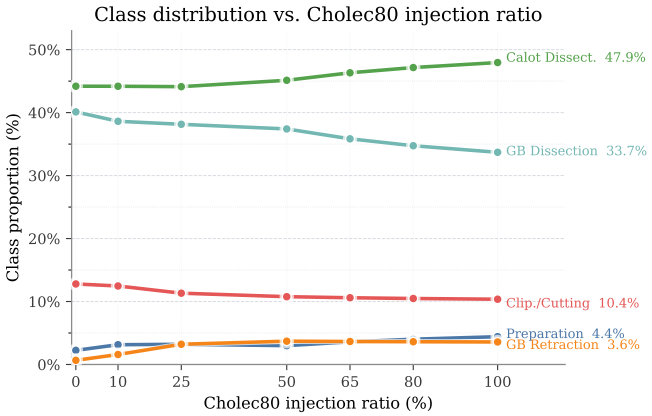

Fig. 3. Cholec80 data injection into the OmniRAS-PR training set (here sampled to have 1 frame per second). (a) Training-set size grows from 19.9k to 95.3k frames. (b) The two minority classes (*Preparation*, *GB Retraction*) are progressively boosted by the more balanced Cholec80 distribution.

reveals whether the domain gap originates primarily in the SSL pretraining stage. If the robotic backbone performs substantially better without needing laparoscopic injection, this confirms that modality-specific pretraining is the most effective strategy; if both backbones converge given enough injection, fine-tuning data diversity alone may compensate for the pretraining mismatch.

IV. Results

A. E1: Laparoscopic Features Do Not Transfer to Robotic Data

TABLE III

E1 – Cross-domain evaluation in both directions (mean $\pm$ std over 3 seeds).

Metric	Fine-tuned on Cholec80 (%)		Fine-tuned on OmniRAS-PR (%)	
	Cholec80 (in)	OmniRAS-PR (cross)	OmniRAS-PR (in)	Cholec80 (cross)
Per-video Accuracy	82.2 ± 1.1	16.8 ± 5.7	82.7 ± 7.4	10.2 ± 0.2
Frame Accuracy	81.9 ± 1.8	16.0 ± 6.4	83.3 ± 8.5	9.2 ± 0.2
F1 (macro)	74.2 ± 1.4	16.9 ± 1.8	65.0 ± 9.0	9.7 ± 0.2
F1 (weighted)	82.4 ± 1.7	18.5 ± 5.2	85.8 ± 2.5	8.3 ± 0.2
Precision (macro)	71.5 ± 2.4	21.1 ± 5.0	73.7 ± 7.9	16.8 ± 0.4
Recall (macro)	78.5 ± 0.3	30.4 ± 6.7	62.7 ± 6.3	21.2 ± 8.2

TABLE IV

E1 – Per-class precision/recall/F1 on OmniRAS-PR (mean $\pm$ std over 3 seeds, in %).

Class	Precision (%)	Recall (%)	F1 (%)
Preparation	34.0 ± 11.3	51.9 ± 21.3	40.2 ± 11.4
GallbladderRetraction	0.5 ± 0.4	50.7 ± 35.5	0.9 ± 0.8
CalotTriangleDissection	52.8 ± 17.9	17.3 ± 3.7	25.3 ± 4.3
ClippingCutting	11.7 ± 14.0	30.9 ± 15.7	16.0 ± 17.0
GallbladderDissection	6.4 ± 7.4	1.5 ± 2.2	2.4 ± 3.4

The EndoViT SPR backbone fine-tuned on Cholec80 reaches 81.9% validation accuracy on Cholec80 itself. When the same checkpoint is evaluated on the OmniRAS-PR robotic test set—without any robotic exposure during training—performance collapses. Table III reports frame-level metrics averaged over three random test splits, each comprising five held-out OmniRAS videos (out of 41). Overall accuracy drops to $16.0 \pm 6.4\%$, macro-F1 to $16.9 \pm 1.8\%$, and mean per-video accuracy to $16.8 \pm 5.7\%$—roughly five-fold degradation

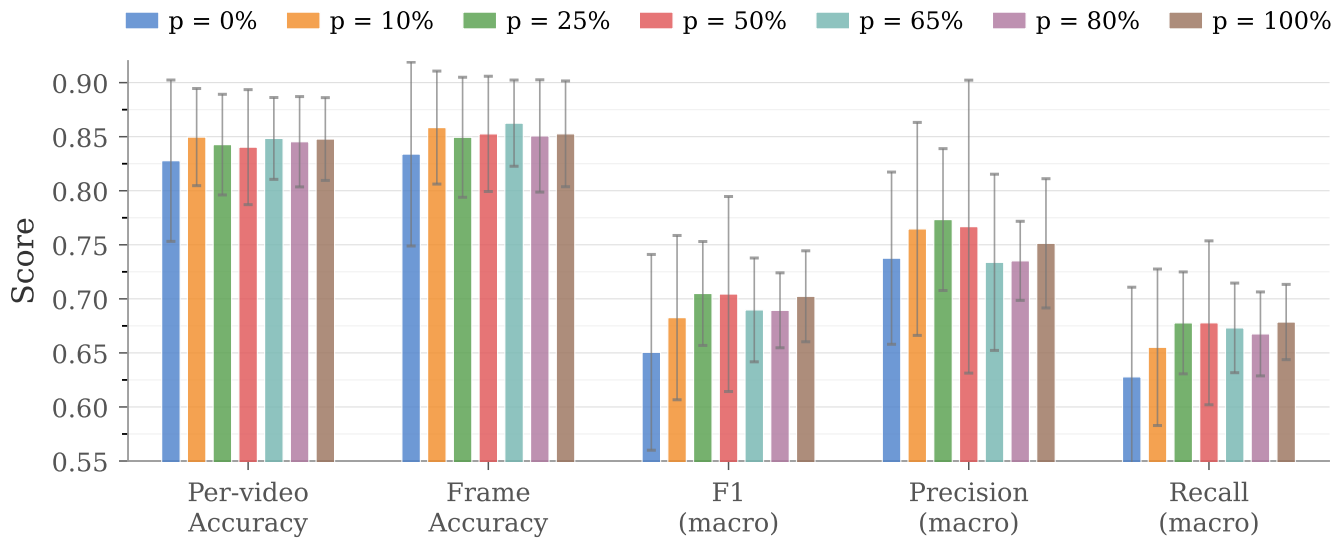

Fig. 4. E3 – Cholec80 injection in curve (Mean ± std over 3 seeds). Each group of bars shows a different evaluation metric; colours denote the fraction p of Cholec80 data injected into the OmniRAS-PR training set. Macro metrics peak at p=25%–50%, dip at p=65%–80%, and partially recover at p=100%, revealing a non-monotonic trade-off between cross-modality diversity and robotic-signal dilution.

relative to the in-domain Cholec80 result. Notably, a majority-class baseline (always predicting *GallbladderDissection*) would achieve ∼40.7% accuracy on the OmniRAS-PR test set, meaning the cross-domain model performs *worse than trivial guessing*—it does not merely lack discriminative power but actively misclassifies the dominant class. We observe a similarly severe degradation in the opposite direction: the robotic-pretrained model fine-tuned on OmniRAS-PR performs strongly in-domain but drops sharply when evaluated on Cholec80 (Table III), where the majority-class baseline (∼37% for *CalotTriangleDissection*) likewise exceeds the cross-domain accuracy of 9.2%. This confirms that the transfer failure is bidirectional and severe.

The per-class breakdown (Table IV) reveals that the transfer failure is highly non-uniform. *Preparation* retains moderate performance (F1 $40.2 \pm 11.4\%$), likely because its visual signature (empty abdomen, trocar insertion) is similar across modalities. *CalotTriangleDissection* and *ClippingCutting* degrade substantially but remain above chance. In contrast, *GallbladderRetraction* (F1 $0.9 \pm 0.8\%$) and *GallbladderDissection* (F1 $2.4 \pm 3.4\%$) are essentially unrecognised, consistent with these phases exhibiting the largest visual difference between laparoscopic and robotic instrument appearances.

We emphasise that the goal of this work is not to maximise absolute phase-recognition accuracy—for which temporal models such as Trans-SVNet [29] and TeCNO [30] achieve >90% on Cholec80—but rather to *quantify* the laparoscopic–robotic representation gap under controlled conditions. Our frame-level evaluation isolates the visual backbone's contribution from temporal smoothing effects, providing a better measure of cross-modality transfer.

B. E2: The Gap Is Deep, Not Shallow

Table V compares three fine-tuning regimes applied to the laparoscopic-pretrained EndoViT on OmniRAS-PR: fully frozen backbone (only the classification head is updated), unfreezing one additional linear layer, and fully unfrozen (all parameters updated). Here we intentionally report only the first fine-tuning step described in the EndoViT protocol (short adaptation stage) so that all regimes are compared under the same constrained setting. The objective is not to maximise absolute performance, but to study the adaptation trend and quantify transferability from laparoscopic to robotic data as trainable capacity increases. This design mirrors standard linear-probing and partial-unfreezing protocols widely used in NLP and vision foundation-model evaluation [13], [14], where constrained adaptation reveals representation quality independently of downstream training budget.

TABLE V

E2 – Frozen vs. unfrozen backbone: EndoViT SPR fine-tuned on OmniRAS-PR with different numbers of trainable layers (mean ± std over 3 runs, in %).

Metric	Frozen (%)	Linear layer (%)	Full (%)
Per-video Accuracy	69.6 ± 5.8	76.6 ± 4.5	$\mathbf{80.1 \pm 6.2}$
Frame Accuracy	69.5 ± 7.1	76.7 ± 7.0	$\mathbf{79.4 \pm 8.9}$
F1 (macro)	41.7 ± 7.6	52.1 ± 8.9	$\mathbf{67.5 \pm 17.0}$
F1 (weighted)	69.1 ± 6.0	77.1 ± 5.7	$\mathbf{80.0 \pm 7.6}$
Precision (macro)	53.3 ± 12.1	59.7 ± 10.8	$\mathbf{72.1 \pm 16.9}$
Recall (macro)	40.0 ± 6.6	50.4 ± 8.0	$\mathbf{66.9 \pm 14.8}$

The results show a clear, monotonic improvement as more layers are unfrozen. Moving from the fully frozen backbone to full fine-tuning raises frame-level accuracy from 69.5% to 79.4% (+9.9 pp) and macro-F1 from 41.7% to 67.5% (+25.4 pp). The largest gains appear in recall (macro), which jumps from 40.0% to 66.9% (+26.9 pp), indicating that the frozen backbone systematically misses minority phases that require robotic-specific features to recognise. Unfreezing a single linear layer already recovers a substantial fraction of the gap (accuracy 76.7%, macro-F1 52.1%), suggesting that

the deepest layers are partially reusable but the intermediate representations still carry a strong laparoscopic bias.

These findings confirm that the domain shift between laparoscopic and robotic surgery is not merely a surface-level classifier mismatch: it extends deep into the visual representation. Adapting only the task head still yields a sizable performance gap relative to full adaptation. motivating the full fine-tuning and robotic pretraining strategies explored in E3 and E4.

C. E3: Progressive Injection Partially Bridges the Gap

Figure 4 reports the effect of progressively injecting Cholec80 laparoscopic data into the OmniRAS-PR training set at seven fractions ($p \in \{0, 10, 25, 50, 65, 80, 100\}\%$). At p=0% (OmniRAS-PR-only), the 4-epoch robotic MAE checkpoint already reaches 83.4% frame-level accuracy after OmniRAS-PR fine-tuning, confirming that even a deliberately non-finalized pretraining stage is adaptable given sufficient in-domain supervision.

We note that with three seeds, individual pairwise differences between adjacent injection fractions may not reach statistical significance. We therefore focus on the *overall trend shape*—an inverted-U followed by partial recovery—rather than on point-to-point comparisons. The consistency of this pattern across multiple metrics (macro-F1, precision, recall) strengthens our confidence that it reflects a genuine phenomenon rather than random variation.

The metrics most sensitive to injection are F1 (macro), precision (macro), and recall (macro), all of which reflect performance on minority phases. These improve strongly from p=0% to the mid-range injections, where macro-F1 peaks at p=25% (70.5%, up from 65.1%) and remains similarly high at p=50% (70.4%). Macro precision also peaks at p=25% (77.3%, up from 73.8%). Even 10% of Cholec80 already provides a substantial boost (macro-F1 rises to 68.3%), indicating that small cross-modality exposures disproportionately help the phases whose visual signature benefits from laparoscopic examples.

Beyond the mid-range, macro performance drops at p=65% and p=80% (macro-F1 ≈69%, macro precision ≈73–74%), then partially recovers at p=100% (macro-F1 70.2%, macro precision 75.1%). This indicates that high injection ratios can over-dilute the robotic signal before recovering slightly at full mixing. Per-video accuracy and frame accuracy remain relatively stable across all fractions (∼83–86%), with the highest frame accuracy at p=65% (86.3%), confirming that the injection effect is concentrated on the minority and cross-modality-sensitive phases.

The key to interpreting these results is the label-composition shift shown in Figure 3: injecting Cholec80 increases the effective proportion of under-represented phases in the mixed training set. This improved exposure to minority labels is reflected most directly in the macro-averaged metrics (especially macro-F1), which weight rare phases more heavily than accuracy and therefore capture the benefits of rebalancing. Overall, this rebalancing effect can be more effective than standard data augmentation, which typically perturbs existing samples without substantially changing the underlying label distribution.

More broadly, this reveals one of the key findings of this paper: there is a "sweet spot" for cross-modality injection where a moderate amount of laparoscopic data improves the model's ability to recognise under-represented robotic phases, yielding noticeable gains in macro-level performance, while higher injection ratios can transiently degrade macro performance before partial recovery.

From a transferability perspective, the non-monotonic behaviour suggests that moderate cross-modality supervision is beneficial even from an intentionally undertrained robotic checkpoint, but that naively mixing large amounts of laparoscopic supervision during robotic adaptation can induce negative transfer in the 65–80% regime. In practice, moderate injection acts as a regulariser that improves robustness on rare robotic phases, while preserving enough in-domain robotic signal to maintain correct robotic-specific appearance modelling.

D. E4: Robotic Pretraining

Figure 5 compares two 14-epoch MAE pretraining settings—robotic-corpus+SurgeNet-LP (mixed-modality) and robotic-corpus-only (robotic-only)—under the same three adaptation regimes used in E2 (Frozen, linear layer, Full). In both settings, performance increases strongly as more layers are trainable. For robotic-corpus+SurgeNet-LP, accuracy/macro-F1 rise from 79.4%/61.3% (Frozen) to 88.3%/74.0% (Full); for robotic-corpus-only, they rise from 78.3%/56.8% to 88.1%/73.9%.

The key to interpreting Figure 5 is that pretraining composition matters most when adaptation capacity is constrained. In Frozen and linear-layer adaptation, robotic-corpus+SurgeNet-LP is consistently stronger on most macro metrics (e.g., macro-F1: 61.3% vs 56.8% in Frozen, 72.4% vs 68.7% in linear), indicating better minority-phase robustness before full adaptation. Once all layers are unfrozen, the two pretraining settings largely converge in overall performance, with a metric-level trade-off: robotic-corpus+SurgeNet-LP is slightly higher on macro precision (76.8% vs 74.4%), while robotic-corpus-only is higher on macro recall (75.9% vs 72.8%) and per-video accuracy (84.9% vs 84.0%).

More broadly, this reveals a complementary effect similar to E3: balanced cross-modality exposure improves robustness where supervision capacity is limited, but modality-matched robotic signal remains important for preserving robotic-specific behaviour under full adaptation. In other words, there is no single global winner across all regimes and metrics.

From a transferability perspective, E4 confirms that modality-aware pretraining substantially closes the laparoscopic-to-robotic gap and reduces sensitivity to the adaptation regime. In practice, mixed robotic-corpus+SurgeNet-LP pretraining is a safer choice when fine-tuning is partially constrained, while with full fine-tuning both pretraining variants achieve similarly strong overall performance with a precision–recall trade-off.

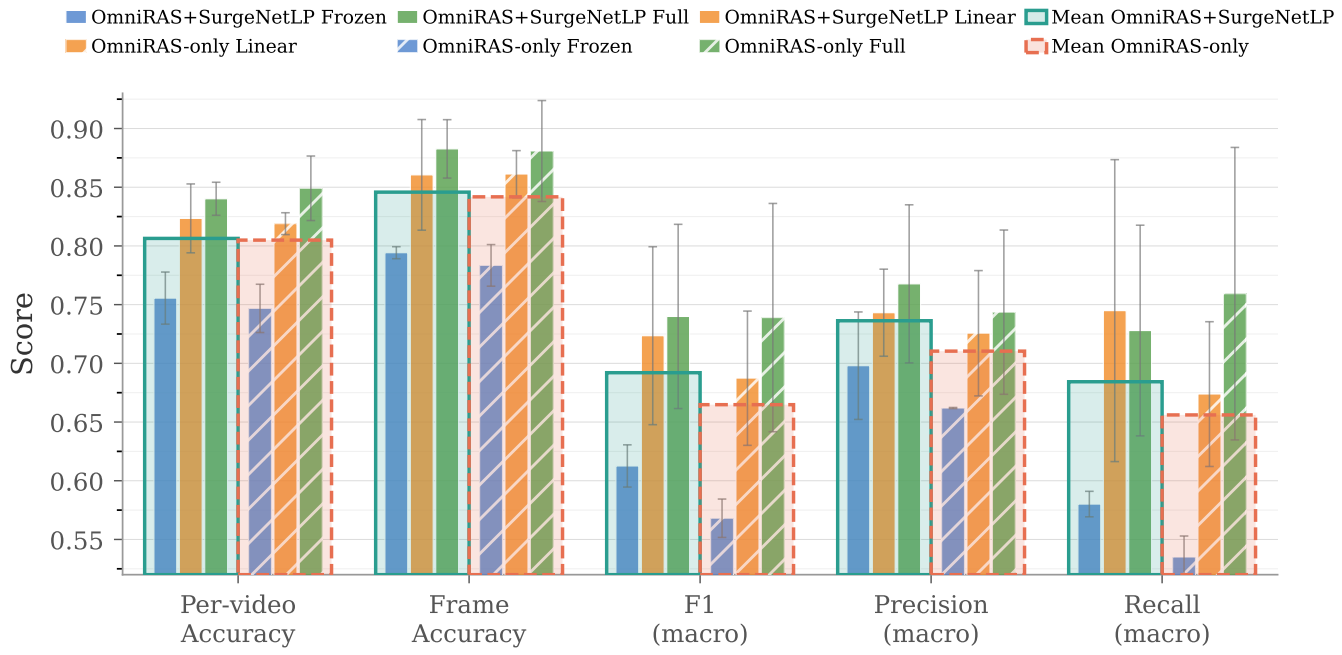

Fig. 5. E4 – Histogram comparison of pretraining settings OmniRAS+SurgeNet-LP and robotic-corpus-only (mean ± std over 3 seeds), where "SurgeNet-LP" denotes only the laparoscopic subset of SurgeNet. Each metric group shows the three adaptation regimes (Frozen, linear layer, Full). Blue horizontal segments mark the mean across Frozen/linear/Full within each metric category for each pretraining setting.

V. Conclusion

We presented the first systematic study of representation transferability between laparoscopic and robotic surgery in the context of self-supervised learning. To support this investigation, we curated one of the largest purely robotic multi-dataset video collections reported in the literature (43,425 clips; 499.4 h; ∼71.3M frames) and introduced a robotic phase-recognition dataset (OmniRAS-PR, 41 videos, 788k frames) mapped to the Cholec80 label space for controlled cross-modality comparison.

Taken together, E1–E4 support four linked conclusions. First, E1 shows that cross-domain transfer degrades sharply in both directions: a laparoscopic pipeline does not generalise reliably to robotic data, and the reverse transfer from robotic to laparoscopic also collapses, confirming a genuine bidirectional modality gap. Second, E2 shows that this gap is representation-deep: even in the short first-step fine-tuning protocol, performance improves monotonically as more backbone capacity is unfrozen, so adaptation cannot be reduced to a classifier-head problem. Third, E3 shows that cross-modality injection helps, but not monotonically: moderate Cholec80 mixing improves macro-level minority-phase robustness, while high mixing ratios can induce temporary negative transfer before partial recovery. Fourth, E4 shows that pretraining composition interacts with adaptation capacity: mixed robotic-corpus+SurgeNet-LP pretraining is safer when fine-tuning is constrained, whereas with full fine-tuning mixed and robotic-only pretraining becomes similarly strong overall but with distinct precision–recall operating points.

This leads to a practical view of transferability as a three-axis design problem: pretraining modality, adaptation depth, and fine-tuning data composition must be chosen jointly. For resource-constrained settings, a modality-aware backbone with at least partial unfreezing is a more robust default than frozen-backbone transfer. For high-capacity settings, full fine-tuning can recover much of the mismatch, after which operating-point choices (precision-oriented vs. recall-oriented behaviour) become more relevant than raw accuracy alone. In the same spirit, moderate cross-modality injection should be treated as a controlled regulariser for rare phases, not as a monotonic substitute for in-domain robotic signal. More broadly, our results suggest that, in surgical vision, maximising backbone-level generalisation in isolation is not always the most important objective. The downstream task often depends on small, highly task-specific regions of the scene, so what matters in practice is how effectively supervised learning can shape the prediction head while simultaneously adapting the backbone through unfreezing. In other words, for focused surgical tasks, transfer performance is governed not only by the generic quality of the pretrained representation, but by the model's ability to re-specialise its attention toward the limited image evidence that is actually decisive for the task.

Our study has limitations: it focuses on one backbone family, one downstream task (phase recognition), and one label harmonisation protocol. Future work should test whether the same transfer dynamics hold for temporal models, and tool- aware and action-level tasks. Promising directions include modality-aware SSL objectives, curriculum-based mixing schedules, and adaptive fine-tuning policies that explicitly optimize minority-phase reliability.

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